

Heewon Lee

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Education

2021 — Current: B.S. in Computer Science

KAIST, Korea | Double major: Physics | Minor: Material Science and Engineering

- **GPA:** 4.10/4.30
- **Relevant Coursework:** Quantum Simulators, Phase Transitions, Thermal Physics, Quantum Mechanics II, Matrix Groups
- **Honors:** Dean's List (Spring 2021, Spring 2023, Fall 2023), Honor Program (Spring 2023 – Current)
- **Anticipated Graduation:** August 2027

May 2024 — November 2025: Republic of Korea Army

Signal Intelligence & Electronic Warfare Operator

- Honorably discharged as Sergeant
- **Concurrent research** via remote guidance from Prof. Sukyoung Ryu (June 2025 – February 2026)

Research Experience

Programming Systems Laboratory, KAIST

Undergraduate Intern, Advisor: Prof. Kihong Heo | June 2023 – April 2024

- Co-developed **UnitCon**, a static-analysis-driven **targeted unit test synthesizer** for Java programs.
- Focused on **test modification**, **bug discovery** in open-source projects, and **integrating static analysis reports** into the enumerative synthesizer pipeline.
- Co-authored research manuscript, culminating in a **Distinguished Paper Award at FSE 2025**.

Independent Research

Advisor: Prof. Sukyoung Ryu | June 2024 – November 2025 (During Military Service)

- Formalized **asymptotic analysis** as a pedagogical **abstract interpretation**.
- Proved soundness via **Galois connection** and developed **interactive tools** for classroom settings.
- Independently researched and remotely advised during **military service**.
- Peer-reviewed and selected to present at the **ACM SIGPLAN POPL 2026 Student Research Competition**.

Publications & Presentations

UnitCon: Synthesizing Targeted Unit Tests for Java Runtime Exceptions. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE), 2025.

S. Jang, Y. Ryou, **H. Lee**, K. Heo

[ACM SIGSOFT Distinguished Paper Award]

Asymptotic Analysis as an Abstract Interpretation. Poster presentation, ACM SIGPLAN POPL 2026 Student Research Competition (SRC).

H. Lee, S. Ryu

Asymptotic Analysis: The Undergraduate's Abstract Interpretation. Poster presentation, Special Interest Group on Programming Languages (SIGPL) Winter School 2026.

H. Lee, S. Ryu

Presented in Korean

Skills

- **Research Interests:** Quantum computation and information theory, simulation of quantum many-body systems, quantum hardware design
- **Languages & Frameworks:** Python, OCaml, Java, Lean 4
- **Quantum Software:** Qiskit, OpenQASM